Jack Markham

**Professional Self Assessment**

Throughout my time at Southern New Hampshire University, I have gained a more in-depth look at the field of computer science and the skills I will need to enter the field as a software engineer. When I first started the program, I had a basic understanding of programming, but over time I became more comfortable working with different languages, solving problems, and developing applications. Completing my coursework and putting together this ePortfolio has helped me recognize how much I have improved and prepared me to pursue a career in software engineering.

One of the most important things I learned was how to approach problems and find solutions. Working with languages such as C++, Java, and Python helped me understand different ways to develop software. I also learned the importance of writing organized and maintainable code. Throughout the program, I frequently had to troubleshoot errors and make improvements to existing projects. These experiences taught me to be patient, think through my problems, and find solutions instead of getting frustrated and giving up.

Communication and collaboration were also important parts of my education. Through class discussions and peer feedback, I learned to explain technical information and consider different perspectives. I realized that software development involves more than just programming because developers also need to work with others. These are skills I hope to continue to improve as I enter the professional field.

My coursework also helped me develop skills in algorithms, data structures, and databases. I learned how different data structures affect program performance and why choosing the right approach matters. I also gained experience working with MongoDB and MySQL to

organize and retrieve information. Outside of my selected ePortfolio projects, I worked on an Android inventory application and full stack web application. This helped me better understand how different parts of a program work together.

Another important area I learned about was security. I became more aware of the importance of validating user input and protecting information. I learned that security should be considered throughout the development process rather than after the program is completed.

For my CS 499 ePortfolio, I selected three projects that demonstrate different areas of computer science. My first artifact comes from CS 340 and focuses on software design and engineering through a Python and MongoDB animal shelter application. My second artifact comes from CS 300 and demonstrates algorithms and structures through a C++ planner that uses a hash table. My third artifact comes from DAD 220 and focuses on database development using MySQL. Each project gave me an opportunity to improve existing work and demonstrate the skills I developed throughout my education.

Overall, completing this ePortfolio has helped me reflect on my progress and better understand my strengths. I feel most comfortable working with C++, Java, and Python, and I am interested in pursuing a career in software engineering. I enjoy developing applications, solving problems, and finding ways to improve existing programs. My experience at SNHU has given me a strong foundation in computer science, and I look forward to continuing to learn and develop my skills.